

Capstone — Recompile Archive

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An AI-augmented archive for Bitcoin technical writing.

Mingjing Zhang · CSE552 · 2026-06-10

Repo

<https://github.com/mingjing-zhang/capstone-recompile-archive> (monorepo: `api/` + `frontend/` + `root` `docker-compose.yml`). Public. `.env` gitignored, contains only `ANTHROPIC_API_KEY` .

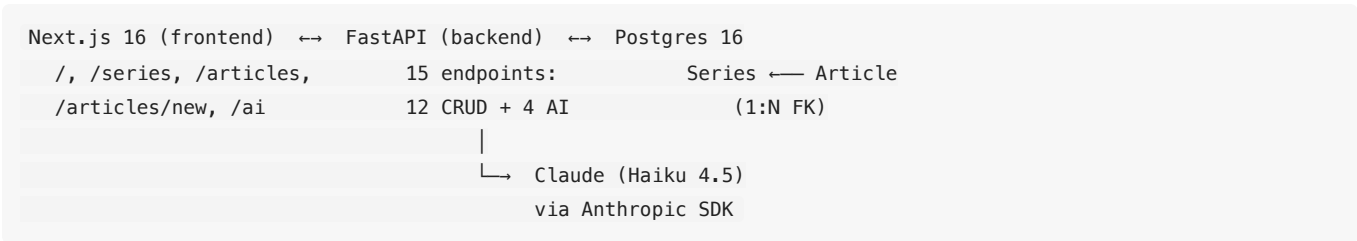
Run locally:

```
git clone https://github.com/mingjing-zhang/capstone-recompile-archive.git
cd capstone-recompile-archive
echo "ANTHROPIC_API_KEY=sk-ant-..." > .env
docker compose up --build
# Frontend → http://localhost:3000
# Backend → http://localhost:8000 (Swagger /docs)
# Seed → docker compose exec backend python seed.py
```

What it is

A personal archive for the kind of long-form Bitcoin protocol writing I do at `@aaron.recompile` on Medium — series like “Not Just HODLing — Real Bitcoin Script Engineering” and “OP_* on Signet — Bitcoin Inquisition”. The CRUD half lets me file articles under series; the AI half makes the archive actually *useful* by letting me search by topic, summarize on demand, classify new drafts, and run multi-step natural-language commands against my own writing.

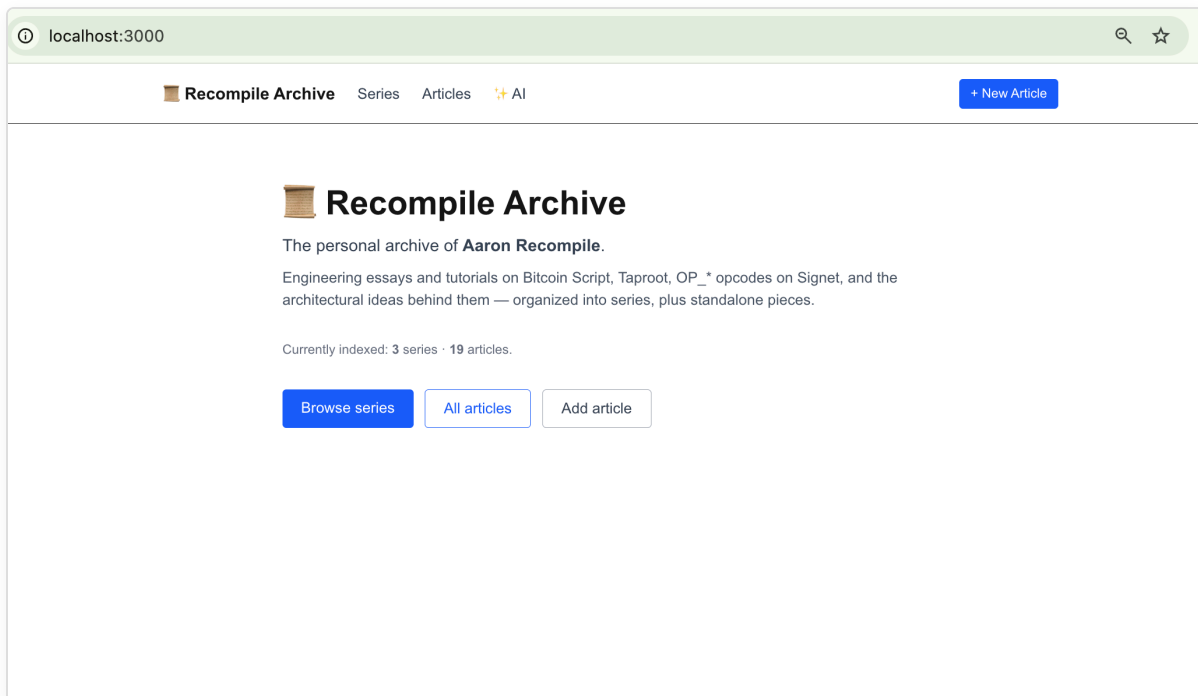
Architecture



- **Two related models** with a real foreign key (`Article.series_id` → `Series.id` , `Series.articles` back-populated, cascade delete).
- **15 endpoints**: 12 standard CRUD across both resources, plus 4 AI endpoints.
- **4 AI endpoints**, each grounded in DB content (not generic LLM chat):
 - `POST /ai/search` — natural-language query → ranked articles with per-match `why_relevant` reasoning.
 - `POST /ai/summarize` — article id → 150–200 word summary + 3–5 key concept tags.
 - `POST /ai/classify` — candidate title/subtitle → suggested series + confidence + alternatives.
 - `POST /ai/agent` — 6-tool agent loop (list/get articles + series, create + update articles), returns `response` , `agent_steps` , `iterations` .
- **Docker**: `api/Dockerfile` , `frontend/Dockerfile` , `root docker-compose.yml` wiring all three services. Postgres healthcheck before backend starts. `ANTHROPIC_API_KEY` interpolated from root `.env` .

Screenshots

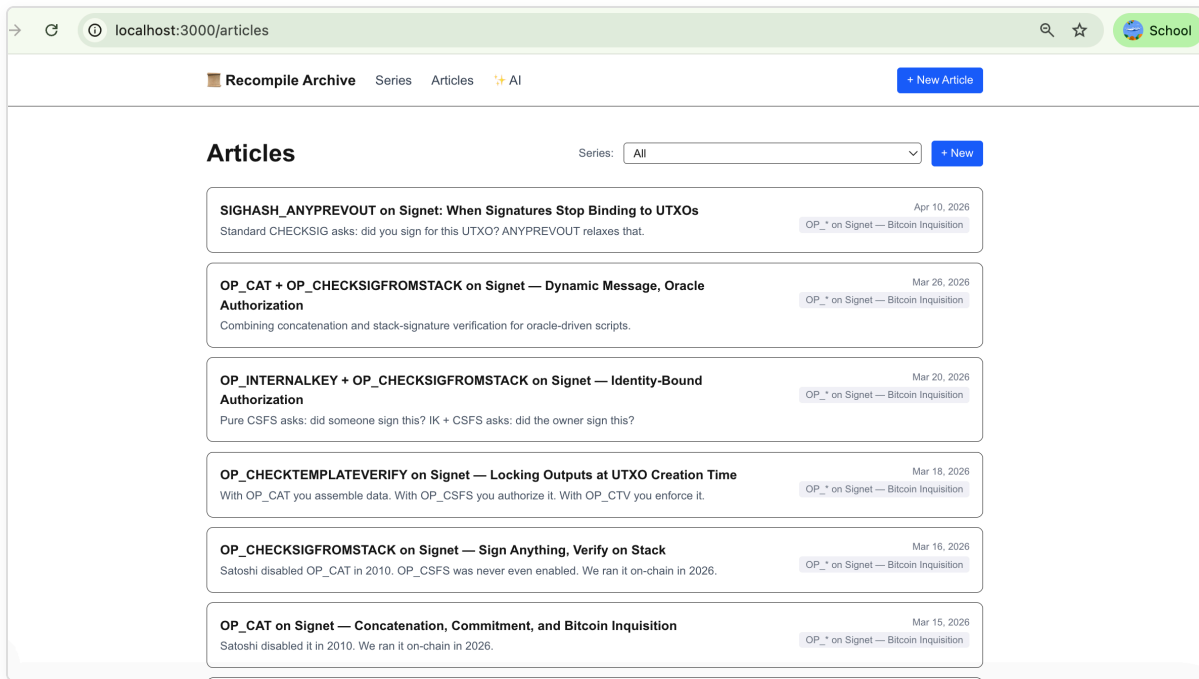
1. Homepage



Homepage — 3 series · 19 articles

2. Articles list (CRUD)

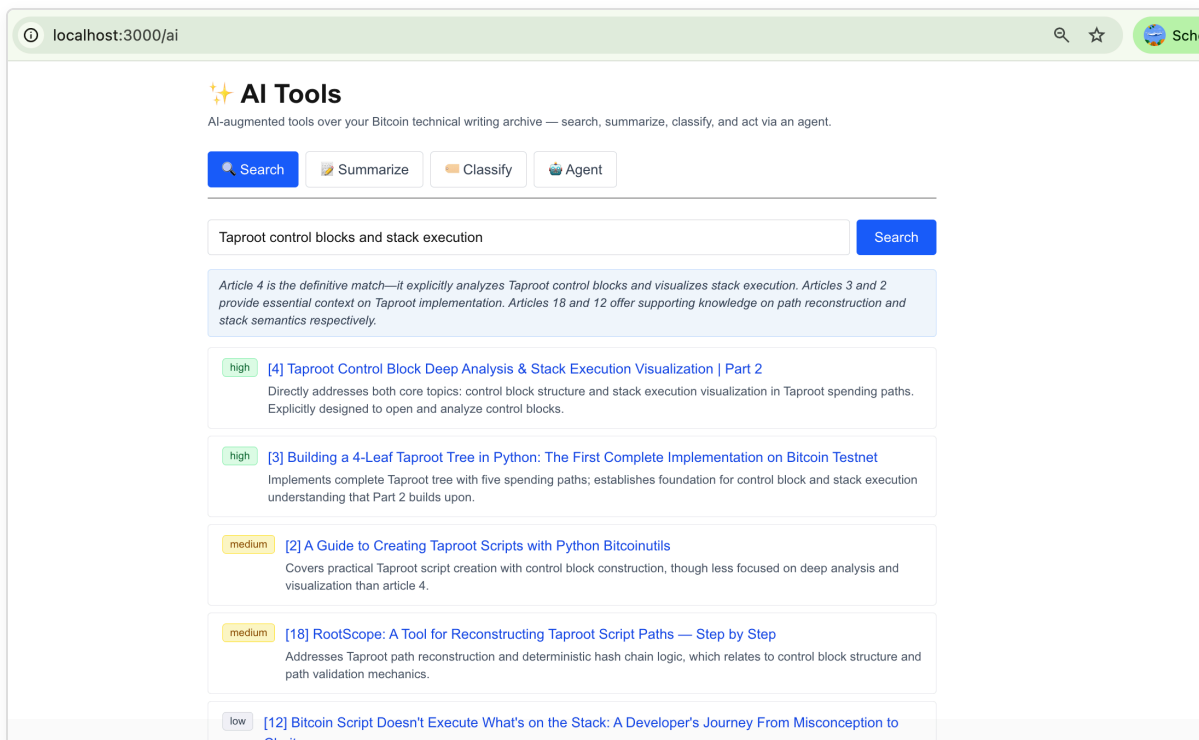
Standard CRUD UI over the `Article` table. Click any title to read; `+ New` to create.



Articles list with 19 real Bitcoin technical articles

3. AI Search

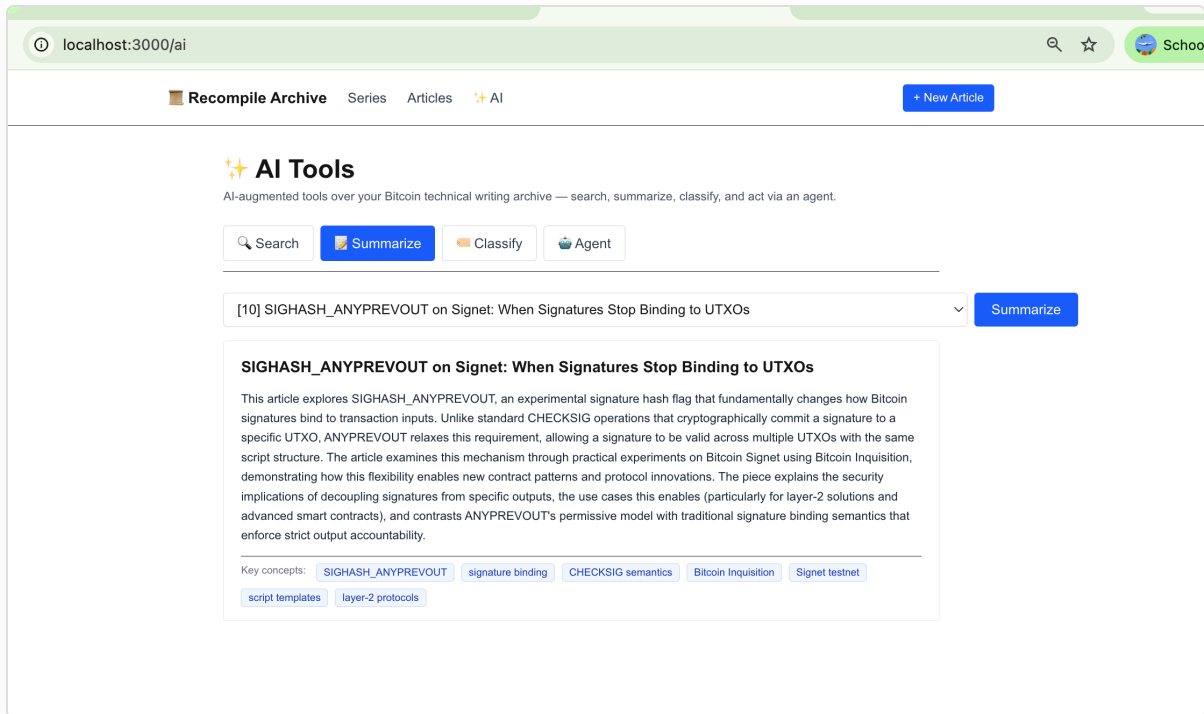
Query: *“Taproot control blocks and stack execution.”* The endpoint returns 5 ranked matches with explicit `relevance` tags (`high` / `medium` / `low`) and a per-result `why_relevant` sentence. Each result links to the underlying article page.



AI search returning 5 ranked Taproot-related articles with reasoning

4. AI Summarize

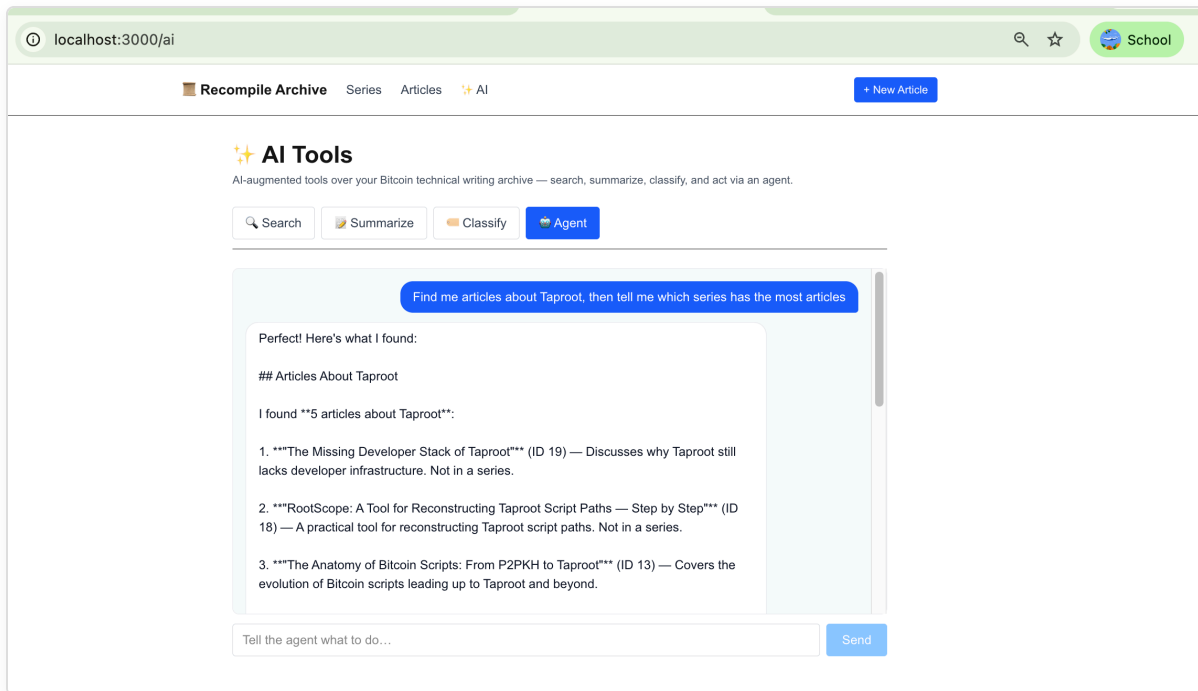
Pick *[10] SIGHASH_ANYPREVOUT on Signet...* from the dropdown → get a 200-word expanded summary plus 8 key concept chips (*SIGHASH_ANYPREVOUT*, *signature binding*, *CHECKSIG semantics*, *Bitcoin Inquisition*, *Signet testnet*, *script templates*, *layer-2 protocols*).



Summarize panel showing generated summary and concept tags

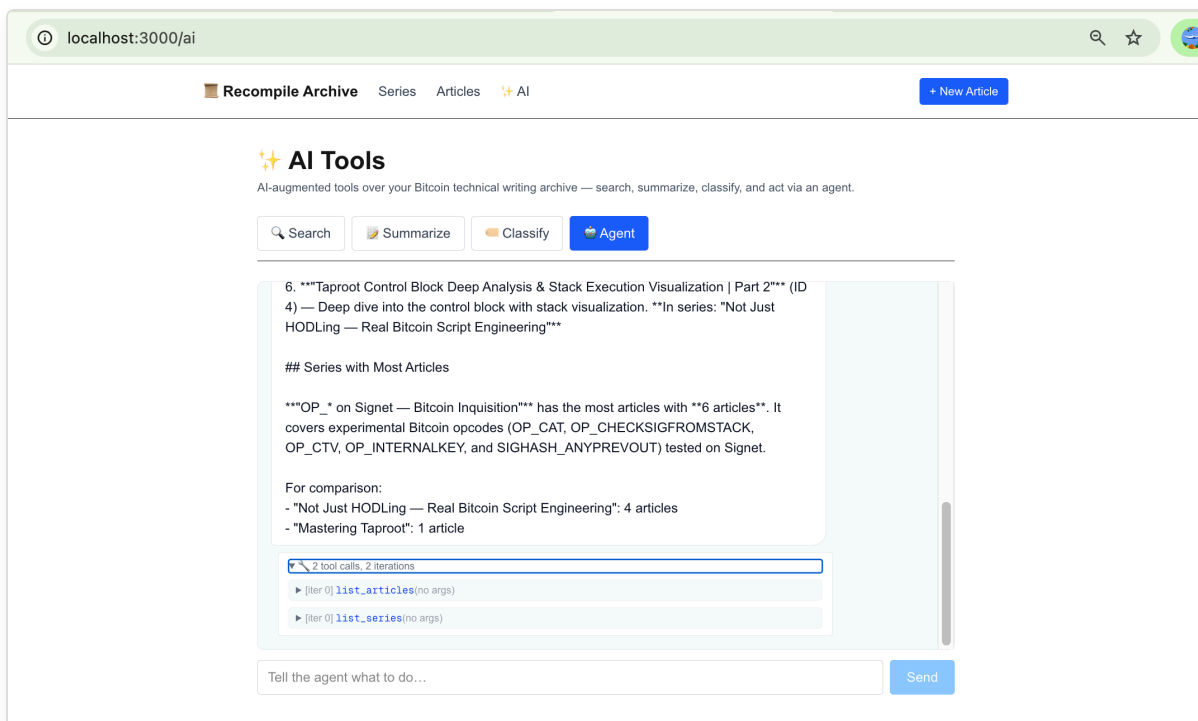
5 + 6. AI Agent (multi-step tool use)

Request: *"Find me articles about Taproot, then tell me which series has the most articles."* The agent calls `list_articles({})` and `list_series({})` in **parallel** during iteration 0, then synthesizes a single answer — listing the 5 matching Taproot articles AND naming *OP_* on Signet — Bitcoin Inquisition* as the largest series (6 articles).



Agent response: 5 Taproot articles found, top series identified

Expanding the `2 tool calls, 2 iterations` accordion reveals the actual tool calls the model made:



Agent tool calls expanded — `list_articles` and `list_series`

This is the same `tool_use` loop pattern from Lab 6, retargeted at `Article / Series` instead of `Book`. The 6 tools are: `list_articles`, `get_article`, `list_series`, `get_series`, `create_article`, `update_article`.

Technical highlight

The 6-tool agent in `api/agent.py` is the centerpiece. It's a vanilla Anthropic `tool_use` loop (max 10 iterations, structured `agent_steps` log), but the tool *descriptions* are doing the heavy lifting — they explicitly tell the model: *“if the user refers to an article by title, look up the id with `list_articles` first; don't fabricate ids.”* That single sentence is what produces the right behavior on Test 4 multi-step requests like the one above. The same lesson from Lab 6 reflection Q3 lands here verbatim: bad tool descriptions don't crash, they cause quiet data corruption.

Grounding, not chat. All four AI endpoints inject real DB content into the system prompt before each call. `/ai/search` lists every article (id + title + subtitle + series) inline in the system prompt; the model can only return ids that actually exist. `/ai/recommend` -style hallucinations of fake book titles (the failure mode I called out in the Lab 5 reflection) are structurally impossible here, because the model is looking at the actual list every turn.

Rubric self-check

Criterion	Pts	Status
Functionality		
Core CRUD works in production	15	✅ Series + Article CRUD via REST + UI
AI feature is meaningful & uses app data	20	✅ 4 endpoints, each grounded in DB; see screenshots 3–6
All major flows work without errors	10	✅ Verified end-to-end (uvicorn + npm dev today; <code>docker compose up --build</code> for live demo)
Technical Quality		
Frontend: Next.js, multi-page, responsive	10	✅ Next.js 16, 5+ routes (<code>/</code> , <code>/series</code> , <code>/series/[id]</code> , <code>/articles</code> , <code>/articles/[id]</code> , <code>/articles/new</code> , <code>/ai</code>)
Backend: FastAPI, 2+ models, organized code	10	✅ FastAPI, <code>Series</code> + <code>Article</code> w/ FK, code split across <code>main.py</code> + <code>agent.py</code> + <code>schemas.py</code>

Criterion	Pts	Status
Database: Postgres, persists across restarts	5	✅ Postgres 16 in Docker; <code>pgdata</code> volume
AI: Claude API integrated thoughtfully	10	✅ 3 structured-output endpoints (JSON schema enforced) + 1 tool-use agent; Haiku 4.5 for cost
Docker: Dockerfiles + compose, starts clean	5	✅ <code>api/Dockerfile</code> , <code>frontend/Dockerfile</code> , <code>root</code> <code>docker-compose.yml</code> with healthcheck on db
Presentation (live in class)		
Live demo via <code>docker compose up</code>	5	🎤 5-min script below
Clear explanation of problem & solution	5	🎤 In script
Technical highlight clearly explained	5	🎤 Agent loop + grounding-not-recall principle
Total	100	

5-minute demo script

Time	Beat
0:00 – 0:30	<i>Problem.</i> I write a lot of long-form Bitcoin protocol articles, scattered across series. When I need to find “where exactly did I cover Taproot control blocks?”, <code>grep</code> -by-title doesn’t scale.
0:30 – 3:30	Live demo via <code>docker compose up --build</code> . Open <code>/articles</code> → browse the archive. Open <code>/ai/search</code> → query “Taproot control blocks” → AI ranks matches and tells me WHY each one is relevant. Switch to Summarize → pick <code>SIGHASH_ANYPREVOUT</code> → 200-word summary + concept tags. Switch to Classify → paste a candidate new-article title → AI tells me which series it belongs in. Switch to Agent → “find

Time	Beat
	<i>Taproot articles and tell me my biggest series"</i> → multi-step tool call, expand the accordion to show the actual <code>list_articles</code> + <code>list_series</code> calls.
3:30 – 4:30	<i>Technical highlight.</i> The agent loop is ~30 lines: same <code>tool_use</code> pattern from Lab 6, but with article/series tools. The real engineering is in the tool <i>descriptions</i> — single sentences like “ <i>if the user mentions a series by name, look up the id with list_series first; don’t fabricate</i> ” are what make the agent safe instead of confident-but-wrong.
4:30 – 5:00	Next. pgvector embeddings over actual article body text (currently I only ground in title/subtitle metadata). Then the same RAG pattern, retargeted at BIPs and papers, becomes my real day-job tool.

Notes

- **Model:** `claude-haiku-4-5` (~\$1/M in + \$5/M out, roughly 3x cheaper than Sonnet for this workload). Total Anthropic spend across all dev + testing: **< \$0.05**.
- **Code reuse:** builds on Mini Project 2 (`Series / Article` CRUD). The 4 AI endpoints + agent + AI UI are new this capstone. Same identity hygiene as prior labs (`mingjing-zhang/` GitHub, `.env` gitignored, no secrets in history).
- **What docker compose up builds:** Postgres 16 (image), backend (Python 3.11 + uvicorn + anthropic + sqlalchemy), frontend (Node 22 + Next.js 16 dev server). All three start with one command.